RAJALAKSHMI ENGINEERING COLLEGE

Heerachandar S 251801099-2025-AIDS

H2

Started on	Saturday, 12 September 2026, 2:03 PM
State	Finished
Completed on	Saturday, 12 September 2026, 2:22 PM
Time taken	18 mins 33 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
int n;
if(scanf ("%d",&n)!=1)
return 0;
if(n<0){
printf("0\n");
return 0;
}
long long *dp=(long long *)calloc(n+1,sizeof(long long));
dp[0]=1;
for(int i=1; i<=n;i++){
if(i >=1){
dp[i]+=dp[i-1];
}
if(i>=3){
dp[i]+=dp[i-3];
}
}
printf("%lld\n",dp[n]);
free(dp);
return 0;
}
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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